

CHAPTER 0: INTRODUCTION

0.1 EULER - BERNOULLI EQUATION [© 1750]

In **classical beam deflection theory** the following assumptions are made: -

1. The material is homogeneous, isotropic, and obeys Hooke's law.
2. The stresses are within the elastic limit.
3. Plane sections before straining remain plane after straining.

Consider a small element of a beam under flexure as shown below:

Let **R** be the radius of curvature (measured from the centre of curvature O, to the neutral axis), and **dθ** be the change in slope between the ends of the considered element.

Consider the fibers AA distance **y** from the NA.

Original length of fibers AA = $R \, d\theta$

Final length of fibers AA = $(R + y) \, d\theta$

Flexural strain = $[(R + y) \, d\theta - R \, d\theta] / R \, d\theta$

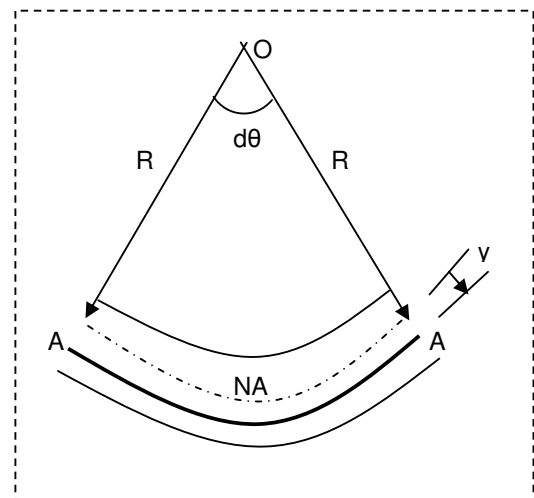
Thus $\epsilon = y/R$

The flexural stress is given by $f = E\epsilon$

Thus $f = Ey/R$

Hence we can write

$$f/y = E/R \text{ ----- Eqn 1}$$



Equilibrium of moments requires that the applied moment be equal and opposite to the moment of resistance of the section, thus

$$M = \int_A y \, f \, dA = \int_A y \, (E/R) \, y \, dA = (E/R) \int_A y^2 \, dA$$

But $\int_A y^2 \, dA = I$, the second moment of area of the section (aka the moment of inertia)

$$\text{Thus } M = (E/R) \, I$$

Hence we can write

$$M/I = E/R \text{ ----- Eqn 2}$$

In Strength of Materials, equations (1) and (2) above are combined to form the classical

Euler - Bernoulli equation expressed as **$f/y = E/R = M/I$**

In theory of structures we shall express:

$$M/EI = 1/R \text{ ----- Eqn 3}$$

The quantity **EI** is the **flexural stiffness** of the beam, while the quantity **1/R** is the **curvature** of the beam. [Note that **R** is the radius of curvature, but **1/R** is the curvature]

0.2 DETERMINATION OF THE CURVATURE

In CHIROMO, any continuous mathematical function is characterized by three quantities at any point $x = a$, namely; -

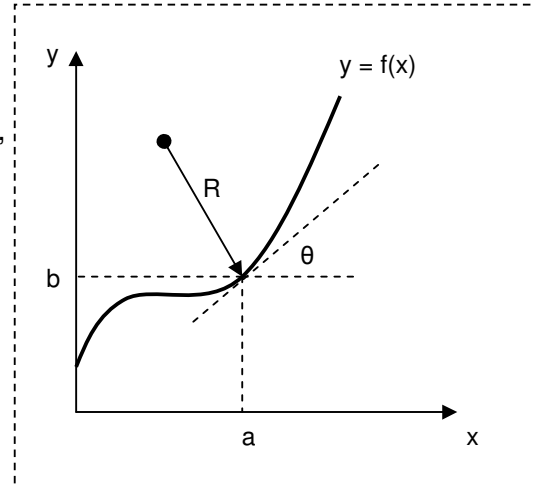
1. The ordinate $y = b$
2. The gradient dy/dx
3. The curvature $1/R$

In engineering, the ordinate is called the **deflection**, and the gradient is called the **slope**.

Thus $y = \delta$ and $dy/dx = \theta$

CHIROMITES believe that

$$\frac{1}{R} = \frac{\frac{d^2y}{dx^2}}{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}^3$$



In structural engineering, the slope is very small compared to unit, thus $dy/dx \ll 1$, and the curvature may be approximated as:

$$1/R = d^2y/dx^2 \text{ ----- Eqn 4}$$

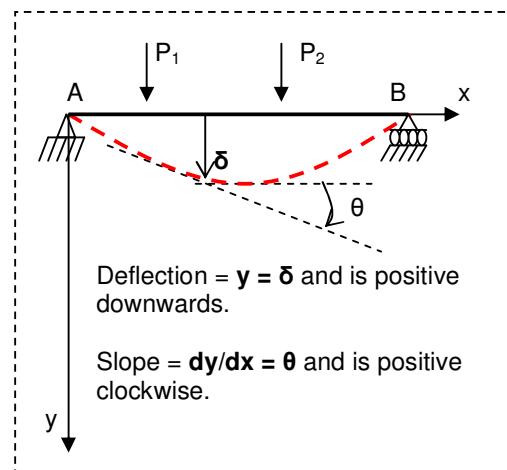
According to the sign convention adopted in FCE 201, positive moment causes sagging, and such moment is plotted below the baseline. This means that a simply supported beam subject to sagging moments deflects downwards as shown below.

In FCE 202 we shall adopt the following sign convention:

1. Deflection is measured positively downwards.
2. Slope is positive if the beam rotates clockwise.
3. The x-y axes system are as shown.

Eqn 3 combined with Eqn 4 can be written as:

$$d^2y/dx^2 = -M/EI \text{ ----- Eqn 5}$$



[The negative sign accounts for our sign convention]

0.3 FROM FCE 201 TO FCE 202

In FCE 201 we expressed $dV/dx = -\omega$ and $dM/dx = V$ so that

$$d^2M/dx^2 = -\omega \quad \text{-----} \quad (a)$$

In FCE 202 we have seen that

$$d^2y/dx^2 = -M/EI \quad \text{-----} \quad (b)$$

Assuming that EI is constant, Eqn (b) can be differentiated twice to obtain:

$$EI d^4y/dx^4 = -d^2M/dx^2 \quad \text{-----} \quad (c)$$

Eqn (a) and Eqn (c) can be combined to relate the displacements directly to the applied loads:

$$EI d^4y/dx^4 = \omega \quad \text{-----} \quad \text{Eqn 6}$$

In theory of structures, Eqn 6 is called the **beam deflection equation**, and forms the bridge on the road from FCE 201 to FCE 202.

Assuming that the loading ω can be expressed as a continuous mathematical function of x , Eqn 6 can be integrated successively four times to obtain mathematical expressions for the slope and deflection profiles

$$EI d^4y/dx^4 = \omega \quad \longrightarrow \quad \text{FBD}$$

$$EI d^3y/dx^3 = \int \omega dx + C_1 \quad \longrightarrow \quad \text{SFD}$$

$$EI d^2y/dx^2 = \int \left\{ \int \omega dx + C_1 \right\} dx + C_2 \quad \longrightarrow \quad \text{BMD}$$

$$EI dy/dx = \int \left\{ \int \left\{ \int \omega dx + C_1 \right\} dx + C_2 \right\} dx + C_3 \quad \longrightarrow \quad \theta\text{-diagram}$$

$$EI y = \int \left\{ \int \left\{ \int \left\{ \int \omega dx + C_1 \right\} dx + C_2 \right\} dx + C_3 \right\} dx + C_4 \quad \longrightarrow \quad \delta\text{-diagram}$$

NOTE

1. It may not be easy to express the applied loads as a continuous mathematical function of x (for example, point load or concentrated moment).
2. There may be no sufficient obvious boundary conditions to solve for C_1 to C_4 .
3. The beam deflection equation is therefore not commonly used to evaluate beam displacements.

0.4 THE DIRECT INTEGRATION METHOD

0.4.1 Curvature, Slope and Deflection

The **direct integration method** (proposed by Daniel Bernoulli © 1750) is based on the assumption that EI is constant, so that the beam curvature can be expressed using Eqn 5

Curvature: $EI \frac{d^2y}{dx^2} = -M$

By integrating twice we obtain:

Slope: $EI \frac{dy}{dx} = \int -M dx + C_1$

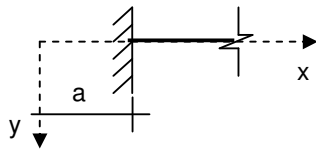
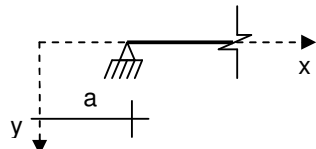
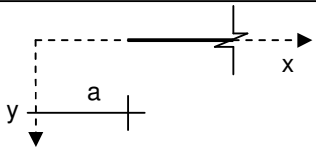
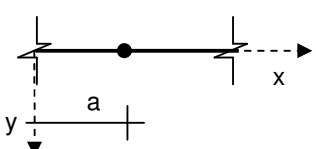
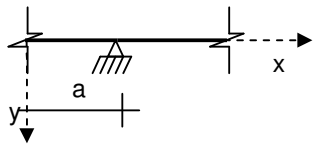
Deflection: $EI y = \int \{ \int -M dx + C_1 \} dx + C_2$

NOTE

1. The moment M is expressed as a function of x , according to FCE 201 Chap 5.
2. The constants of integration C_1 and C_2 are obtained from the existing boundary conditions.

0.4.2 Boundary Conditions

The various boundary conditions are summarized in the table below.

Fixed Support		Deflection = 0 Slope = 0	$y _a = 0$ $\frac{dy}{dx} _a = 0$
Pin support or Roller support at end		Deflection = 0 Moment = 0	$y _a = 0$ $\frac{d^2y}{dx^2} _a = 0$
Free end		Moment = 0 Shear = 0	$\frac{d^2y}{dx^2} _a = 0$ $\frac{d^3y}{dx^3} _a = 0$
Internal hinge		Moment = 0	$\frac{d^2y}{dx^2} _a = 0$
Continuous Support		Deflection = 0	$y _a = 0$

0.4.3 Solution Procedure

The solution procedure may be outlined as follows: -

1. Determine the support reaction components to obtain a **fbd** of the given beam.
2. For each region:
 - a) Consider a **fbd** from the **LH** end to some point, distance **x**, within that region.
 - b) Express the bending moment **M** in terms of **x** according to FCE 201 Chap 5
 - c) Use Eqn 5 to express **$EI \frac{d^2y}{dx^2} = -M$**
 - d) Integrate twice, assuming **EI = constant**.
3. Next region
4. Use the existing boundary conditions to solve for the constants of integration.
5. Obtain mathematical expressions for the **deflection profile** and the **slope profile**.
6. Use these expressions to evaluate the required displacements.

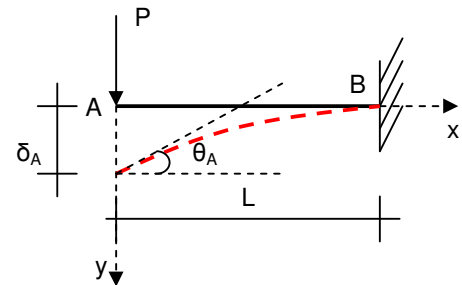
Note

1. The sign convention with regard to bending moment is as agreed in FCE 201.
2. The bending moment diagram is continuous (at least piece-wise) within each region.

0.4.4 Worked Examples

Example 1

Determine the deflection δ_A and slope θ_A of the free end A, in terms of P, L and EI. Assume EI = constant.



Solution

$$M = -Px$$

$$EI \frac{d^2y}{dx^2} = Px$$

$$EI \frac{dy}{dx} = Px^2/2 + C_1$$

$$EI y = Px^3/6 + C_1x + C_2$$

Boundary conditions

$$1. \frac{dy}{dx} \Big|_L = 0 \Rightarrow PL^2/2 + C_1 = 0 \Rightarrow C_1 = -PL^2/2$$

$$2. y \Big|_L = 0 \Rightarrow PL^3/6 + C_1L + C_2 = 0 \Rightarrow C_2 = PL^3/3$$

$$\text{Deflection profile: } EI \delta = Px^3/6 - PL^2x/2 + PL^3/3$$

$$\text{Slope profile: } EI \theta = Px^2/2 - PL^2/2$$

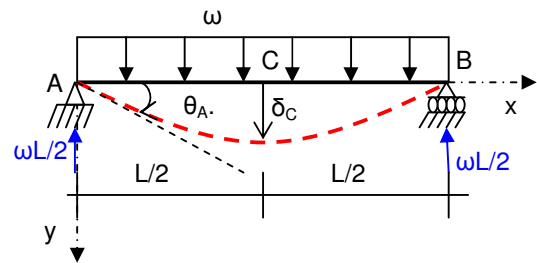
$$\text{At A: } x = 0: EI \delta_A = PL^3/3 \Rightarrow \delta_A = PL^3/3EI$$

$$\text{And } EI \theta_A = -PL^2/2 \Rightarrow \theta_A = -PL^2/2EI$$

Example 2

Determine the deflection at midspan δ_C , and the slope at A, θ_A , in terms of ω , L and EI.

Assume EI = constant.



Solution

$$M = \omega Lx/2 - \omega x^2/2$$

$$EI \frac{d^2y}{dx^2} = -\omega Lx/2 + \omega x^2/2$$

$$EI \frac{dy}{dx} = -\omega Lx^2/4 + \omega x^3/6 + C_1$$

$$EI y = -\omega Lx^3/12 + \omega x^4/24 + C_1x + C_2$$

Boundary conditions

$$1. \quad y|_0 = 0 \Rightarrow C_2 = 0$$

$$2. \quad y|_L = 0 \Rightarrow C_1 = \omega L^3/24$$

Deflection profile: $EI \delta = -\omega Lx^3/12 + \omega x^4/24 + \omega L^3x/24$

Slope profile: $EI \theta = -\omega Lx^2/4 + \omega x^3/6 + \omega L^3/24$

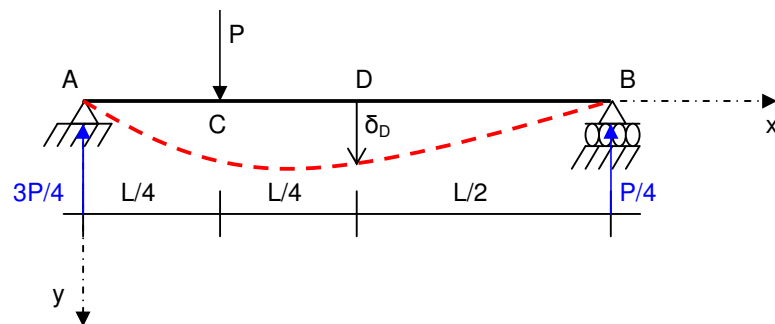
At midspan, $x = L/2$: $EI \delta_C = -\omega L(L/2)^3/12 + \omega(L/2)^4/24 + \omega L^3(L/2)/24$

$$\Rightarrow \delta_C = 5\omega L^4/384EI$$

At A, $x = 0$: $EI \theta_A = \omega L^3/24 \Rightarrow \theta_A = \omega L^3/24EI$

Example 3

Determine the deflection at midspan in terms of P, L and EI. Take EI = constant.



Solution

Region AC: $0 \leq x \leq L/4$

$$M = 3Px/4$$

$$EI \frac{d^2y}{dx^2} = -3Px/4$$

$$EI \frac{dy}{dx} = -3Px^2/8 + C_1$$

$$EI y = -3Px^3/24 + C_1x + C_2$$

Region CB: $L/4 \leq x \leq L$

$$M = P(L-x)/4$$

$$EI \frac{d^2y}{dx^2} = P(x-L)/4$$

$$EI \frac{dy}{dx} = P(x^2/2 - Lx)/4 + C_3$$

$$EI y = P(x^3/6 - Lx^2/2)/4 + C_3x + C_4$$

Boundary conditions:

1. Region AC: $y|_0 = 0 \Rightarrow C_2 = 0$ ----- (i)

2. Region CB: $y|_L = 0 \Rightarrow -PL^3/12 + C_3L + C_4 = 0$ ----- (ii)

3. Continuity at C: $\delta_{C(L)} = \delta_{C(R)}$

$\Rightarrow -3P(L/4)^3/24 + C_1L/4 + C_2 = P(L^3/384 - L^3/32)/4 + C_3L/4 + C_4$ ----- (iii)

4. Continuity at C: $\theta_{C(L)} = \theta_{C(R)}$

$\Rightarrow -3P(L/4)^2/8 + C_1 = P(L^2/32 - L^2/4)/4 + C_3$ ----- (iv)

Solving eqns (i) to (iv) simultaneously gives:

$$C_1 = 7PL^2/128, \quad C_2 = 0, \quad C_3 = 11PL^2/128, \quad C_4 = -PL^3/384$$

Using the deflection equation of region CB (where midspan falls), we have

$$\begin{aligned} EI \delta_D &= P((L/2)^3/6 - L(L/2)^2/2)/4 + 11PL^2/128 (L/2) - PL^3/384 \\ &= -5PL^3/192 + 11PL^3/256 - PL^3/384 \\ &= 11PL^3/768 \end{aligned}$$

Hence, at midspan $\delta_D = 11PL^3/768EI$

NOTE

1. The direct integration method, as outlined above, is quite efficient for beams made up of one region only. (See examples 1 and 2 above)
2. However the direct integration method becomes tedious and too involving if the BMD is piece-wise continuous. In each region there will be two constants of integration to be solved for! (See example 3). This can lead to a large set of equations.
3. This apparent difficulty was overcome by **Macaulay**, as discussed in Chapter 1.

Welcome to
FCE 202
